

LiDAR

synthetic-scrap-pit-v1

, , ,

. versioned JSON 3 .

Artifact fingerprint SHA-256

be782bd9e5ffdd86a94c545da9d673ce206d76 956fabdbad20a8974a1 199 120b

1.

, , LiDAR 2 , 5 .

		SHA-256
examples/environment.v1.json	sensor	9d1253e0ed80a3c2b67 609c7397a499bf8e6061cbd7144ff273 732e9aa32a33a
examples/generator.v2.json	, seed	33d21c149 410f69e4ac1dcb2fc52caab2aaaffd61a8f4e986fb165ae740efd9f
examples/quality-profile.v1.json	Sensor quality	a89f33e158c937e727d45 523ad103a4b349 298 041dce1f12 630fce6f2b6ee91c

JSON .JSON fingerprint , DOCX PDF .

2.

world sensor scan 2 . World , sensor sensor scan .

World XYZ	m, deg	, +Z	, , stream
Sensor polar	mm, mdeg	sensor	gRPC ScanFrame

```
Sensor .
direction(angle) = cos(angle) * u0 + sin(angle) * u90
world_point = p0 + distance * direction(angle)
Sensor scan plane, , 0 deg 4 .
scan_plane(a, b) = p0 + a * u0 + b * u90
rotation_axis = u0 x u90
H0 = p0 + ((floor_z - p0.z) / u0.z) * u0
floor_measurement_line(t) = H0 + t * u90
```

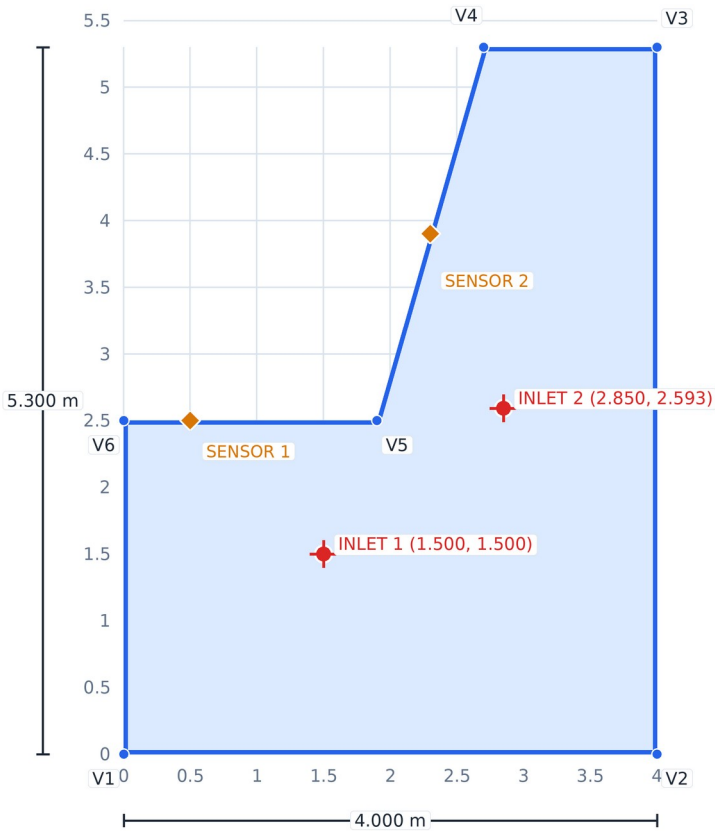
World V1 (0, 0, 0) .XY Z . degree angle 0 angle 90 u0 u90 .

3.

Environment ID	synthetic-scrap-pit-v1		
X	0.000 - 4.000 m	bounding	
Y	0.000 - 5.300 m	bounding	
Z	0.000 m		
Z	10.000 m		
	10.000 m	Z - Z	
	14.760 m2	polygon	
	147.600 m3	x	
	World XY m		
V1	(0, 0)	V1 -> V2	4.000000 m
V2	(4, 0)	V2 -> V3	5.300000 m
V3	(4, 5.3)	V3 -> V4	1.300000 m
V4	(2.7, 5.3)	V4 -> V5	2.912044 m
V5	(1.9, 2.5)	V5 -> V6	1.900000 m
V6	(0, 2.5)	V6 -> V1	2.500000 m

PLAN VIEW - WORLD XY

unit: m | right-handed, Z up



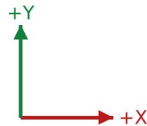
LEGEND

Storage boundary

Sensor XY projection

Inlet XY position

Exact coordinates: Markdown specification



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be782bd9e5ffdd86a94c545da9d673ce
206d76956fabdbad20a8974a1199120b

1. World XY , .

4.

XY

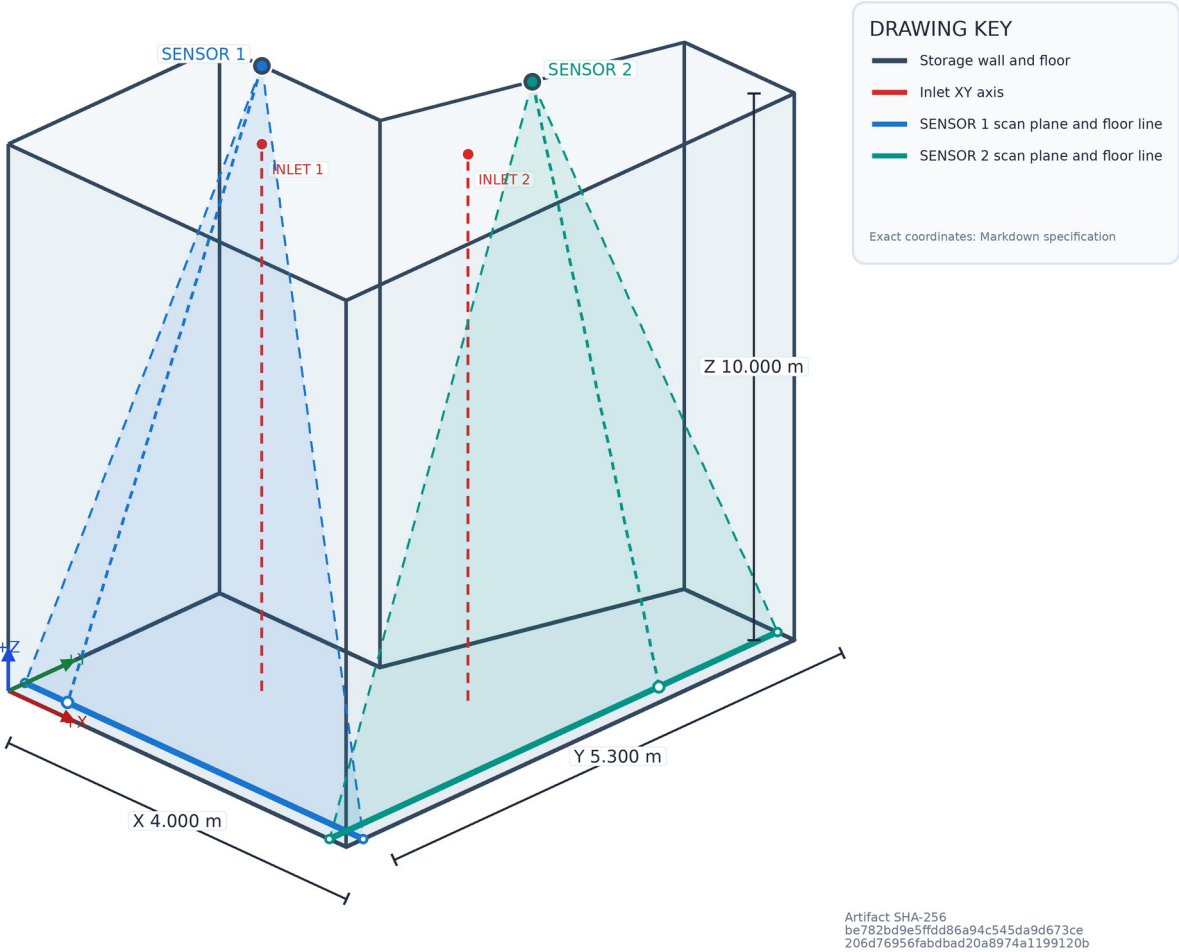
XY

World X m			World Y m		index	
INLET 1		1.5	1.5		0	
INLET 2		2.85	2.593		1	
Sensor	p0 m	u0	u90	deg		0 deg m
lidar_1	[0.5, 2.5, 10]	[0, -0.224147 692 921 427, -0.9745 551 866 149]	[-1, 0, 0]	12.952765		[0.5, 0.2, 0]
lidar_2	[2.3, 3.9, 10]	[0.148340 452 930 245, 0, -0.988936 352 868 298]	[0, -1, 0]	8.530766		[3.8, 3.9, 0]
Sensor	u0 x u90			A m		B m
lidar_1	[0, 0.9745 551 866 149, -0.224147 692 921 427]		H0 + t * [-1, 0, 0]	[4, 0.2, 0]		[0, 0.2, 0]
lidar_2	[-0.988936 352 868 298, 0, -0.148340 452 930 245]		H0 + t * [0, -1, 0]	[3.8, 5.3, 0]		[3.8, 0, 0]

p0 sensor u0 u90 scan plane u0 x u90 360 deg scan plane floor_z_m
, sensor

ISOMETRIC VIEW - WORLD XYZ

orthographic technical projection | unit: m | empty-floor reference



2. , sensor .

5.

	86 400.000 s	24
	0.8 - 1.2	
	0.5 - 1.5	
	300 - 900 s	
	0.85 - 0.95	
	0.0333333 - 0.05	
	0.3 - 1.7	
	60 - 180 s	
cell	0.250 m	
node shape	Y 23 x X 17	node
	0.500 s	
	0.500 m	
	-0.2 - 0.2 m	signed peak
	0.1 - 0.3 m	
	35.000 deg	
	32	
	0.500	
	0.250 m	
	0.500 m	

Y-major polygon bounding box . Y 5.500 m polygon node .

6. LiDAR

	Sensor 1	Sensor 2
Sample rate	32 000.000 point/s	64 000.000 point/s
Rotation rate	10.000 scan/s	20.000 scan/s
	3200.000	wire
	0.050 - 30.000 m	
noise	0.010 m	
noise	0.030 m	

	enabled	0.500 event/s	0.025 - 0.1 m, 0.5 - 2	0.05 - 0.2 s
	enabled	0.030	0.015 - 0.06 m, 0.05 - 0.3 m, 0.100 m	60 - 300 s
	enabled	20 - 40 s	0.15 - 0.5 m, 0.5 - 2	2 - 8 s
	enabled	0.001000	0.5 - 2 m	point
Sensor dropout	disabled	120 - 240 s	sensor	2 - 5 s

Sensor	HQ	HQ	Wire
lidar_1	48:1, 80:3	0:2, 24:1	HQ quality >> 2
lidar_2	48:1, 80:3	0:2, 24:1	HQ quality >> 2

7.

	World XYZ, meter, degree
	polygon, floor Z top Z
Sensor	p0, u0 u90 scan plane
	World XY Z
	World XY Z

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